

Assignment 7: Short Baseline Sensing Systems

Due: Monday, December 1

The objective of this assignment is to design and develop a software program to perform simulation, analysis and estimation of the geolocation of a single target in the active-sensing mode.

An active short baseline sensing device consists of one centered transmitter and two receivers, one at each side. The distance from the transmitter to a receiver is D . (Thus, the entire length of the device is $2D$.)

A single target, as a point reflector, is located within the *target region*. The distance from the transmitter to the target is r_0 , the distance from the target to the first receiver is r_1 and the distance to the second receiver is r_2 .

A sequence of N step-frequency *FMCW* waveforms is transmitted, corresponding to the operating frequencies

$$f_n = [(n + N/2)/N] f_0 \quad n = 0, 1, 2, \dots, N-1$$

where f_0 is the center/operating frequency of the sequence. The frequency span is from $0.5f_0$ to $1.5f_0$, which results in a bandwidth of f_0 . If the space-domain version is preferred, the corresponding wavelengths can be described in the form

$$\lambda_n = [N/(n + N/2)] \lambda_0 \quad n = 0, 1, 2, \dots, N-1$$

You can also model the data-acquisition procedure with a sequence of coherent probing functions

$$p_n(t) = \exp(-j2\pi f_n t)$$

The N -point data sequence collected at the location of the first receiver is

$$g_1(n) = A \exp(j2\pi f_n \tau_1) = A \exp(j2\pi (r_0 + r_1)/\lambda_n)$$

where v is the propagating speed and τ_1 is the time delay due to the propagation, from the transmitter to the target then to the first receiver, where $\tau_1 = (r_0 + r_1)/v$. Similarly, at the second receiver, the data sequence is in the form

$$g_2(n) = A \exp(j2\pi f_n \tau_2) = A \exp(j2\pi (r_0 + r_2)/\lambda_n)$$

where $\tau_2 = (r_0 + r_2)/v$.

Given these two N -point sequences $\{g_1(n)\}$ and $\{g_2(n)\}$, design and implement simple techniques to estimate the location of the target and then evaluate the performance by assessing the estimation errors and resolution.

Apply backward propagation algorithm (bistatic version) for the construction of the target profile as the intersection of two bands ellipsoids. A N -frame video can be constructed to illustrate the convergence of the geolocation task.

Suggestions of software design and development:

1. Functionalities of the program

- (a) Your program will allow you to select the location of a point reflector within the “target region”
- (b) Your program will then generate two sequences of complex data samples, $\{g_1(n)\}$ and $\{g_2(n)\}$, at the two receiver positions
- (c) The program will then perform image reconstruction to estimate the location of the target reflector

2. Design variables

- (a) Operating frequency f_0 or wavelength λ_0
- (b) Bandwidth B
- (c) Number of frequencies/wavelengths N
- (d) Separation of the transmitter-receiver D (Formulate D in terms of wavelength λ_0 for simplicity)
- (e) Range distance of the target location r_0
- (f) Bearing angle of the target location θ_0

3. System performance evaluation parameters

- (a) Accuracy of the target location
- (b) Target resolution

Report:

1. Cover sheet
2. Final image of the target profile
3. Video sequence
4. Discuss the accuracy and effectiveness of these algorithms with respect to (1) separation distance, (2) frequency bandwidth, (3) range distance, and (4) bearing angle of the target location.
5. Matlab code
6. Summary